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High electron mobility transistor and method for fabricating the same

Abstract

A method for fabricating a high electron mobility transistor (HEMT) includes the steps of forming a buffer layer on a substrate, forming a barrier layer on the buffer layer, forming a p-type semiconductor layer on the barrier layer, forming a hole injection buffer layer (HIBL) on the p-type semiconductor layer, and forming a gate electrode on the HIBL.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to a high electron mobility transistor (HEMT) and fabrication method thereof.

2. Description of the Prior Art

(2) High electron mobility transistor (HEMT) fabricated from GaN-based materials have various advantages in electrical, mechanical, and chemical aspects of the field. For instance, advantages including wide band gap, high break down voltage, high electron mobility, high elastic modulus, high piezoelectric and piezoresistive coefficients, and chemical inertness. All of these advantages allow GaN-based materials to be used in numerous applications including high intensity light emitting diodes (LEDs), power switching devices, regulators, battery protectors, display panel drivers, and communication devices.

SUMMARY OF THE INVENTION

(3) According to an embodiment of the present invention, a method for fabricating a high electron mobility transistor (HEMT) includes the steps of forming a buffer layer on a substrate, forming a barrier layer on the buffer layer, forming a p-type semiconductor layer on the barrier layer, forming a hole injection buffer layer (HIBL) on the p-type semiconductor layer, and forming a gate electrode on the HIBL.

(4) According to another aspect of the present invention, a high electron mobility transistor (HEMT) includes a buffer layer on a substrate, a barrier layer on the buffer layer, a p-type

semiconductor layer on the barrier layer, a hole injection buffer layer (HIBL) on the p-type semiconductor layer, and a gate electrode on the HIBL.

(5) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1-3 illustrate a method for fabricating a HEMT according to an embodiment of the present invention.

DETAILED DESCRIPTION

(2) Referring to FIGS. 1-3, FIGS. 1-3 illustrate a method for fabricating a HEMT according to an embodiment of the present invention. As shown in FIG. 1, a substrate **12** such as a substrate made from silicon, silicon carbide, or aluminum oxide (or also referred to as sapphire) is provided, in which the substrate **12** could be a single-layered substrate, a multi-layered substrate, gradient substrate, or combination thereof. According to other embodiment of the present invention, the substrate **12** could also include a silicon-on-insulator (SOI) substrate.

(3) Next, a selective nucleation layer (not shown) and a buffer layer **14** are formed on the substrate **12**. According to an embodiment of the present invention, the nucleation layer preferably includes aluminum nitride (AlN) and the buffer layer **14** is preferably made of III-V semiconductors such as gallium nitride (GaN), in which a thickness of the buffer layer **14** could be between 0.5 microns to 10 microns. According to an embodiment of the present invention, the formation of the buffer layer **14** on the substrate **12** could be accomplished by a molecular-beam epitaxy (MBE) process, a metal organic chemical vapor deposition (MOCVD) process, a chemical vapor deposition (CVD) process, a hydride vapor phase epitaxy (HVPE) process, or combination thereof.

(4) Next, a selective unintentionally doped (UID) buffer layer (not shown) could be formed on the surface of the buffer layer **14**. In this embodiment, the UID buffer layer is preferably made of III-V semiconductors such as gallium nitride (GaN) or more specifically unintentionally doped GaN. According to an embodiment of the present invention, the formation of the UID buffer layer on the buffer layer **14** could be accomplished by a molecular-beam epitaxy (MBE) process, a metal organic chemical vapor deposition (MOCVD) process, a chemical vapor deposition (CVD) process, a hydride vapor phase epitaxy (HVPE) process, or combination thereof.

(5) Next, a barrier layer **16** is formed on the surface of the buffer layer **14** or UID buffer layer. In this embodiment, the barrier layer **16** is preferably made of III-V semiconductor such as n-type or n-graded aluminum gallium nitride ($\text{Al}_{\text{sub}.x}\text{Ga}_{\text{sub}.1-x}\text{N}$), in which $0 < x < 1$, the barrier layer **16** preferably includes an epitaxial layer formed through epitaxial growth process, and the barrier layer **16** could include dopants such as silicon or germanium. Similar to the buffer layer **14**, the formation of the barrier layer **16** could be accomplished by a molecular-beam epitaxy (MBE) process, a metal organic chemical vapor deposition (MOCVD) process, a chemical vapor deposition (CVD) process, a hydride vapor phase epitaxy (HVPE) process, or combination thereof.

(6) Next, a p-type semiconductor layer **18** is formed on the barrier layer **16**, a photo-etching process is conducted to pattern or remove part of the p-type semiconductor layer **18**, a passivation layer **20** is formed on the p-type semiconductor layer **18**, another photo-etching process is conducted to pattern or remove part of the passivation layer **20** for exposing the p-type semiconductor layer **18** surface, a silicon layer or more specifically an amorphous silicon layer **22** is formed on the surface of the p-type semiconductor layer **18**, and a gate electrode **24** is formed on the amorphous silicon layer **22**. Preferably, the formation of the amorphous silicon layer **22** and gate electrode **24** could be accomplished by sequentially forming an amorphous silicon layer **22** and a gate electrode **24** on the

passivation layer **20** and the exposed p-type semiconductor layer **18** entirely, and then using a photo-etching process to remove part of the gate electrode **24** and part of the amorphous silicon layer **22** for forming a patterned amorphous silicon layer **22** and a patterned gate electrode **24**, in which the sidewalls of the patterned amorphous silicon layer **22** and gate electrode **24** are aligned with the sidewall of the p-type semiconductor layer **18** underneath.

(7) In this embodiment, the p-type semiconductor layer **18** is a III-V compound semiconductor layer preferably including p-type GaN (pGaN) and the formation of the p-type semiconductor layer **18** on the barrier layer **16** could be accomplished by a molecular-beam epitaxy (MBE) process, a metal organic chemical vapor deposition (MOCVD) process, a chemical vapor deposition (CVD) process, a hydride vapor phase epitaxy (HVPE) process, or combination thereof.

(8) Even though the passivation layer **20** in this embodiment pertains to be a single-layered structure, according to other embodiment of the present invention, it would also be desirable to form a passivation layer **20** made from a dual layer or tri-layer structure, in which the passivation layer **20** could include dielectric material including but not limited to for example silicon oxide, silicon nitride, or aluminum oxide. Moreover, the thickness of the amorphous silicon layer **22** is between $\frac{1}{6}$ to $\frac{1}{10}$ of the entire thickness of the p-type semiconductor layer **18**. For instance, the thickness of the amorphous silicon layer **22** at this stage is preferably between 2-10 nm or most preferably 5 nm while the thickness of the p-type semiconductor layer **18** is preferably between 60-100 nm.

(9) Next, as shown in FIG. 2, an anneal process **26** is conducted to drive silicon atoms from the amorphous silicon layer **22** downward into the p-type semiconductor layer **18** for forming a hole injection buffer layer (HIBL) **28** made of silicon, in which the silicon atoms within the HIBL **28** could include a gradient concentration. For instance, the silicon concentration closer to the bottom surface of the HIBL **28** is slightly less than the silicon concentration closer to the bottom surface of the HIBL **28**. In this embodiment, the anneal process **26** could include a rapid thermal anneal (RTP) process or a furnace anneal process, in which the temperature of the RTP process is between 500-600° C. and the temperature of the furnace anneal process is between 200-600° C.

(10) Preferably, the thickness of the HIBL **28** after being treated with anneal process **26** is slightly greater than the thickness of the amorphous silicon layer **22** formed previously. For instance, the overall thickness of the HIBL **28** is about $\frac{1}{3}$ or most preferably between $\frac{1}{2}$ to $\frac{1}{3}$ of the thickness of the p-type semiconductor layer **18**. It should be noted that if the amorphous silicon layer **22** were directly treated with the anneal process **26** without any covering or any layer on top, the silicon nature of layer **22** would be easily oxidized into silicon oxide. To prevent this, the present invention first forms the amorphous silicon layer **22** and the gate electrode **24** on the surface of the p-type semiconductor layer **18** and then conducts an anneal process **26** to transform the amorphous silicon layer **22** into the HIBL **28**.

(11) Next, as shown in FIG. 3, a photo-etching process is conducted to remove part of the passivation layer **20** adjacent to two sides of the gate electrode **24** for forming two openings (not shown), and then form conductive materials into the openings along with additional photo-etching process for forming a source electrode **30** and a drain electrode **32** adjacent to two sides of the gate electrode **24**. In this embodiment, the gate electrode **24**, the source electrode **30**, and the drain electrode **32** are preferably made of metal, in which the gate electrode **24** is preferably made of Schottky metal while the source electrode **30** and the drain electrode **32** are preferably made of ohmic contact metals. According to an embodiment of the present invention, each of the gate electrode **24**, source electrode **30**, and drain electrode **32** could include gold (Au), Silver (Ag), platinum (Pt), titanium (Ti), aluminum (Al), tungsten (W), palladium (Pd), or combination thereof. Preferably, it would be desirable to conduct an electroplating process, sputtering process, resistance heating evaporation process, electron beam evaporation process, physical vapor deposition (PVD) process, chemical vapor deposition (CVD) process, or combination thereof to form electrode materials in the aforementioned openings, and then pattern the electrode materials through one or

more etching processes to form the gate electrode **24**, source electrode **30**, and the drain electrode **32**. This completes the fabrication of a HEMT according to an embodiment of the present invention.

(12) Overall, the present invention first forms an amorphous silicon layer on the surface of a patterned p-type semiconductor layer, forms a gate electrode on the amorphous silicon layer, and then conducts an anneal process to drive silicon atoms from the amorphous silicon layer into the p-type semiconductor layer underneath for forming a HIBL. According to a preferred embodiment of the present invention, the silicon atoms within the HIBL could be used as donors to neutralize acceptors such as magnesium (Mg) on surface of the p-type semiconductor layer so that it would more difficult for holes to enter the p-type semiconductor layer and the barrier layer thereby reducing gate leakage. Moreover, HIBL could also be used to improve reliability test for high temperature gate bias (HTGB) of the HEMT device and increase V_g operating range so that more flexibility could be provided to the designers on circuit design.

(13) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A method for fabricating a high electron mobility transistor (HEMT), comprising: forming a buffer layer on a substrate; forming a barrier layer on the buffer layer; forming a p-type semiconductor layer on the barrier layer; patterning the p-type semiconductor layer; forming a passivation layer on the p-type semiconductor layer; patterning the passivation layer to expose the p-type semiconductor layer; forming a silicon layer on the p-type semiconductor layer; forming a gate electrode on the silicon layer; performing an anneal process to transform the silicon layer into a hole injection buffer layer (HIBL); and forming a source electrode and a drain electrode adjacent to two sides of the gate electrode.
 2. The method of claim 1, wherein the silicon layer comprises an amorphous silicon layer.
 3. The method of claim 1, wherein the anneal process comprises a rapid thermal anneal process.
 4. The method of claim 1, wherein a thickness of the silicon layer is less than a thickness of the HIBL.
 5. The method of claim 1, wherein the buffer layer comprises gallium nitride (GaN).
 6. The method of claim 1, wherein the barrier layer comprise $\text{Al}_{\text{sub}.x}\text{Ga}_{\text{sub}.1-x}\text{N}$.
 7. The method of claim 1, wherein the p-type semiconductor layer comprises p-type gallium nitride (pGaN).
 8. The method of claim 1, wherein the HIBL comprises a gradient concentration of silicon.
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